Knights problem

How can a knight jump on an NxN chessboard in such a way that it visits every square exactly once? Consider two cases:

1.The knight's tour ends on the same square where it begun
2.The knight's tour ends on a dfferent square

Cryptarithmetics

1.SEND + MORE = MONEY
2.TWO + TWO = FOUR

Einstein zebra problem (three houses)

1.There are three houses, three nationals, three animals, three cigarettes.
2.The Englishman lives in the first house on the left.
3.In the house immediately on the right of that housing the wolf, they smoke Luky Strike.
4.The Spaniard smokes Kent.
5.The Russian has a horse.

Who smokes LM? Who has the dog?

Einstein zebra problem (five houses)

1.There are five houses.
2.The Englishman lives in the red house.
3.The Spaniard owns the dog.
4.Coffee is drunk in the green house.
5.The Ukrainian drinks tea.
6.The green house is immediately to the right of the ivory house.
7.The Old Gold smoker owns snails.
8.Kools are smoked in the yellow house.
9.Milk is drunk in the middle house.
10.The Norwegian lives in the first house.
11.The man who smokes Chesterfields lives in the house next to the man with the fox.
12.Kools are smoked in the house next to the house where the horse is kept.
13.The Lucky Strike smoker drinks orange juice.
14.The Japanese smokes Parliaments.
15.The Norwegian lives next to the blue house.

Who drinks water? Who owns the zebra?

Maps colouring problem

Is there a way to color a map using N colors such that no adjacent countries do have the same color? Find such a map coloring.

Crosswords

Read an empty crosswords square (i.e. an MxN square and the positions of the black cells). And read a dictionary of words. Fill the crosswords square, such that all words formed on lines and columns appear in the dictionary.

Magic square

Fill an NxN square with all numbers from 1 to N^2 such that the sum of all elements on any line, any column and any diagonal are all the same.

Sudoku

Solve a sudoku game.

1.A 9x9 board, divided in 9 quares of 3x3.
2.Each cell has a number from 1 to 9.
3.Any line has all numbers exactly once.
4.Any column has all numbers exactly once.
5.Any square has all numbers exactly once.

The board is read partially filled. It has to be filled completely.

Timetable

The following items are available:

- A list of classes;
- A list of classrooms;
- A list of tutors, and the list of subjects they can teach;
- Lists of subjects to be taught to classes;

Decide on the classes timetable (i.e. who teaches what, when, where and to whom).

Car sequencing

The car sequencing problem involves scheduling the assembly of cars which are distinguished into classes by different sets of options. The assembly line has different stations where the various options (air-conditioning, sun-roof, radio) are installed by teams. Each such station has a certain workload capacity given by the parameters M and N, denoting M out of a sequence of N cars can be fitted the given option. The objective is to produce such a sequence of cars that (i) the demand for each car class is satisfied and (ii) none of the capacity constraints is violated.

Job-shop scheduling

The following are given:

- A finite set of n jobs
- Each job i consists of a chain of n_i operations
- A finite set of m machines
- Each machine can handle at most one operation at a time
- Each operation needs to be processed during an uninterrupted period of a given length on a given machine

The purpose is to find a schedule, that is, an allocation of the operations to time intervals to machines, that has minimal length.

Balanced Academic Curriculum

Design a balanced academic curriculum by assigning periods to courses in a way that the academic load of each period is balanced, i.e., as similar as possible. The curriculum must obey the following administrative and academic regulations:

1.Academic curriculum: an academic curriculum is defined by a set of courses and a set of prerequisite relationships among them.
2.Number of periods: courses must be assigned within a maximum number of academic periods.
3.Academic load: each course has associated a number of credits or units that represent the academic effort required to successfully follow it.
4.Prerequisites: some courses can have other courses as prerequisites.
5.Minimum academic load: a minimum amount of academic credits per period is required to consider a student as full time.
6.Maximum academic load: a maximum amount of academic credits per period is allowed in order to avoid overload.
7.Minimum number of courses: a minimum number of courses per period is required to consider a student as full time.
8.Maximum number of courses: a maximum number of courses per period is allowed in order to avoid overload.
The goal is to assign a period to every course in a way that the minimum and maximum academic load for each period, the minimum and maximum number of courses for each period, and the prerequisite relationships are satisfied. An optimal balanced curriculum minimises the maximum academic load for all periods. Note that we could consider other types of

balance criterion such as minimising the sum of the academic load of all periods.

Dinosaur Mystery

- Skeletons of Eucentrosaurus, Hadrosaurus, Herrerasaurus, Megasaurus, and Nuoerosaurus have been found in Argentina, Canada, China, England, and the U.S.
- None of these dinosaurs has been found in more than one of these countries.
- The dinosaur from China is the largest of the bunch.
- Herrerasaurus was the smallest.
- Megasaurus was larger than Hadrosaurus or Eucentrosaurus.
- The dinosaurs from China and North America were plant-eaters.
- Megasaurus ate meat.
- Eucentrosaurus lived in North America.
- The dinosaurs from North America and England lived later than Herrerasaurus.
- Hadrosaurus lived in Argentina or the U.S.

Which dinosaur lived in which country?

	Eucentrosaurus	**Hadrosaurus**	**Herrerasaurus**	**Megasaurus**	**Nuoerosaurus**
Argentina	.	.	.	.	.
Canada	.	.	.	.	.
China	.	.	.	.	.
England	.	.	.	.	.
United States	.	.	.	.	.

Making Widgets

- To make a widget using a widget-maker, you must put a metal blank in the slot, push the button, turn the crank, pull the lever, and turn the key, but these actions must be done in a certain order.
- The first thing you do is either pull the lever or turn the key.
- You can't put the blank in the slot until after you push the button and turn the crank.
- You have to do something after you put in the blank.
- The key won't turn when there is no blank in the slot.
- You must push the button immediately after you pull the lever.

How do you make a widget?

	Blank in slot	**Push button**	**Turn crank**	**Pull lever**	**Turn key**
First	.	.	.	.	.
Second	.	.	.	.	.
Third	.	.	.	.	.
Fourth	.	.	.	.	.
Fifth	.	.	.	.	.

Murder Mystery

One of the five men committed a murder on March 17 (St. Patrick's Day.) Each made one false statement and three true statements. Who is the murderer?

Otto	I was in Chicago.	I'm innocent.	The Kid did it .	I know now Mickey.
Curly	I'm innocent.	Never owned gun	The Kid knows me.	I was in Detroit.
Slim	Curly owns a gun.	The murder was on St. Patrick's day.	Otto was in Chicago.	One of us did it.
Mickey	I'm innocent.	The Kid was never in Pontiac.	I don't know Otto.	Curly & me were in Detroit.
The Kid	I'm innocent.	I've never been in Pontiac .	I don't know Curly.	Otto erred when he said I did it.

Scheduling Problem

Classes are offered periods 1-8 with 1-4 in the morning and 5-8 in the afternoon.

- You need three classes on Monday and you can use MATH, CHEM, PHIL, or HIST.
- CHEM is offered odd periods and MATH is offered all periods.
- HIST is offered periods 2 and 4. PHIL is offered periods 5 and 7.

- CHEM and MATH are taught on south campus.
- HIST and PHIL are taught on north campus.
- You can't schedule back-to-back classes on opposite ends of campus.

You have to work mornings or afternoons and you don't want a first period class. Can you satisfy all these constraints? If so, do you tell your boss you will work in the mornings or in the afternoons?

	MATH	CHEM	PHIL	HIST
1				
2				
3				
4				
5				
6				
7				
8				

The Dogs

Mr. Adams, Mrs. Barber, Mr. Cox, Miss Duke, and Miss Evans live in a row of houses in the same block. Each owns a dog. The dogs they own are a beagle, a collie, a dachshund, a poodle, and a retriever.

- Mr. Cox and Miss Duke live at the ends of the row of houses.
- A woman owns the retriever.
- The collie lives in the middle house.
- Miss Duke does not own the poodle.
- The dachshund was a gift from the owner's husband.
- The retriever lives between the collie and the beagle.
- Which breed of dog does each person own?

	beagle	**collie**	**dachshund**	**poodle**	**retriever**
Mr. Adams	.	.	.	.	.
Mrs. Barber	.	.	.	.	.
Mr. Cox	.	.	.	.	.
Miss Duke	.	.	.	.	.
Miss Evans	.	.	.	.	.

The Train

On a certain train, the crew consists of the brakeman, the fireman, and the engineer. Their names in alphabetical order are Jones, Robinson, and Smith. On the train are also three passengers with corresponding names.

- Mr. Robinson lives in Detroit.
- The brakeman lives halfway between Detroit and Chicago.
- Mr. Jones earns exactly $20,000 a year.
- Smith once beat the fireman at billiards.
- The brakeman's next-door neighbor, one of the passengers, earns exactly three times as much as the brakeman.
- The passenger living in Chicago has the same name as the brakeman. What is the engineer's name?

	brakeman	**fireman**	**engineer**
Jones	.	.	.
Robinson	.	.	.
Smith	.	.	.

	Detroit	**between**	**Chicago**
Mr. Jones	.	.	.
Mr. Robinson	.	.	.
Mr. Smith	.	.	.